Week 5: Customer Payment Transaction Management System

Scenario:

The e-commerce enterprise needs a structured database system to record and oversee payment activity for customer orders. The system should accurately log transaction details, monitor real-time payment statuses, handle both cleared and rejected transactions, and generate payment-related analytical reports for business audits.

Students will design and implement a Payment Processing Module seamlessly integrated with the existing Customer and Order Management modules.

Tasks to be Done:

1. Create Payment Table

Payment Entity

Stores customer payment transaction records. Attribute	**Description**
TXN_ID	Unique payment identifier
ORDER_REF	Order reference key
TXN_DATE	Date of transaction
PAY_METHOD	UPI/Card/Net Banking/Cash
TXN_STATUS	Success/Failed/Pending
AMOUNT_PAID	Paid transaction amount

```
SQL> CREATE TABLE ORDER_PAYMENTS(
  2  TXN_ID VARCHAR2(7),
  3  ORDER_REF VARCHAR2(7),
  4  TXN_DATE DATE,
  5  PAY_METHOD VARCHAR2(15),
  6  TXN_STATUS VARCHAR2(10),
  7  AMOUNT_PAID NUMBER(10,2),
  8  PRIMARY KEY (TXN_ID),
  9  FOREIGN KEY(ORDER_REF) REFERENCES Orders(ORDER_REF));

Table created.
```

3. Store Payment Details

```
SQL> INSERT INTO ORDER_PAYMENTS
  2  (TXN_ID, ORDER_REF, TXN_DATE, PAY_METHOD, TXN_STATUS, AMOUNT_PAID)
  3  VALUES
  4  ('TXN0001', 'ORD0001', DATE '2026-01-10', 'Credit Card', 'Completed', 55850.00);

1 row created.

SQL>
SQL> INSERT INTO ORDER_PAYMENTS
  2  (TXN_ID, ORDER_REF, TXN_DATE, PAY_METHOD, TXN_STATUS, AMOUNT_PAID)
  3  VALUES
  4  ('TXN0002', 'ORD0002', DATE '2026-01-25', 'UPI', 'Completed', 3350.00);

1 row created.

SQL>
SQL> INSERT INTO ORDER_PAYMENTS
  2  (TXN_ID, ORDER_REF, TXN_DATE, PAY_METHOD, TXN_STATUS, AMOUNT_PAID)
  3  VALUES
  4  ('TXN0003', 'ORD0004', DATE '2026-02-18', 'Debit Card', 'Completed', 8300.00);

1 row created.

SQL>
SQL> INSERT INTO ORDER_PAYMENTS
  2  (TXN_ID, ORDER_REF, TXN_DATE, PAY_METHOD, TXN_STATUS, AMOUNT_PAID)
  3  VALUES
  4  ('TXN0004', 'ORD0006', DATE '2026-03-15', 'Net Banking', 'Completed', 7000.00);

1 row created.
```

```
SQL> SELECT * FROM ORDER_PAYMENTS;

TXN_ID  ORDER_R TXN_DATE  PAY_METHOD      TXN_STATUS AMOUNT_PAID
------- ------- --------- --------------- ---------- -----------
TXN0001 ORD0001 10-JAN-26 Credit Card     Completed        55850
TXN0002 ORD0002 25-JAN-26 UPI             Completed         3350
TXN0003 ORD0004 18-FEB-26 Debit Card      Completed         8300
TXN0004 ORD0006 15-MAR-26 Net Banking     Completed         7000
TXN0006 ORD0010 18-MAY-26 UPI             Completed         5200
TXN0007 ORD0012 20-JUN-26 Credit Card     Completed         2450
TXN0008 ORD0014 15-AUG-26 Debit Card      Completed        28000
TXN0009 ORD0014 14-AUG-26 Debit Card      Failed           28000
TXN0005 ORD0008 20-APR-26 COD             Completed         5700

9 rows selected.
```

4. Manage Successful and Failed Transactions

Tasks:

Write SQL queries to:

1. Display all successful payments.
2. Find failed transactions.
3. Count total successful and failed payments.

```
SQL> SELECT * FROM ORDER_PAYMENTS WHERE TXN_STATUS = 'Completed';

TXN_ID  ORDER_R TXN_DATE  PAY_METHOD      TXN_STATUS AMOUNT_PAID
------- ------- --------- --------------- ---------- -----------
TXN0001 ORD0001 10-JAN-26 Credit Card     Completed        55850
TXN0002 ORD0002 25-JAN-26 UPI             Completed         3350
TXN0003 ORD0004 18-FEB-26 Debit Card      Completed         8300
TXN0004 ORD0006 15-MAR-26 Net Banking     Completed         7000
TXN0006 ORD0010 18-MAY-26 UPI             Completed         5200
TXN0007 ORD0012 20-JUN-26 Credit Card     Completed         2450
TXN0008 ORD0014 15-AUG-26 Debit Card      Completed        28000
TXN0005 ORD0008 20-APR-26 COD             Completed         5700

8 rows selected.
```

```
SQL> SELECT * FROM ORDER_PAYMENTS WHERE TXN_STATUS = 'Failed';

TXN_ID  ORDER_R TXN_DATE  PAY_METHOD      TXN_STATUS AMOUNT_PAID
------- ------- --------- --------------- ---------- -----------
TXN0009 ORD0014 14-AUG-26 Debit Card      Failed           28000
```

```
SQL> SELECT TXN_STATUS, COUNT(TXN_STATUS) FROM ORDER_PAYMENTS GROUP BY TXN_STATUS
  2  ;

TXN_STATUS COUNT(TXN_STATUS)
---------- -----------------
Completed                 8
Failed                    1
```

5. Analyze Payment Methods Used by Customers

Generate payment analysis reports.

Report 1: Payment Mode Analysis

Find:

- Number of transactions using UPI.
- Number of card payments.
- Most preferred payment method.

```
SQL> SELECT COUNT(*) AS UPI_TRANSACTIONS
  2  FROM ORDER_PAYMENTS
  3  WHERE PAY_METHOD = 'UPI';

UPI_TRANSACTIONS
----------------
               2
```

```
SQL> SELECT COUNT(*) AS CARD_PAYMENTS
  2  FROM ORDER_PAYMENTS
  3  WHERE PAY_METHOD IN ('Credit Card', 'Debit Card');

CARD_PAYMENTS
-------------
            5
```

```
SQL> SELECT
  2       PAY_METHOD,
  3       COUNT(*) AS TRANSACTION_COUNT
  4  FROM ORDER_PAYMENTS
  5  GROUP BY PAY_METHOD
  6  ORDER BY TRANSACTION_COUNT DESC;

PAY_METHOD           TRANSACTION_COUNT
-------------------- -----------------
Debit Card                           3
UPI                                  2
Credit Card                          2
COD                                  1
Net Banking                          1
```

```
SQL> SELECT PAY_METHOD, COUNT(*) AS TRANSACTION_COUNT
  2  FROM ORDER_PAYMENTS
  3  GROUP BY PAY_METHOD
  4  ORDER BY TRANSACTION_COUNT DESC
  5  FETCH FIRST 1 ROW ONLY;

PAY_METHOD           TRANSACTION_COUNT
-------------------- -----------------
Debit Card                           3
```

Report 2: Revenue Analysis

Find:

- Total revenue generated.
- Revenue by payment method.
- Average transaction amount.

```
SQL> SELECT
  2       SUM(AMOUNT_PAID) AS TOTAL_REVENUE
  3  FROM ORDER_PAYMENTS
  4  WHERE TXN_STATUS = 'Completed';

TOTAL_REVENUE
-------------
       115850
```

```
SQL> SELECT
  2       PAY_METHOD,
  3       SUM(AMOUNT_PAID) AS TOTAL_REVENUE
  4  FROM ORDER_PAYMENTS
  5  WHERE TXN_STATUS = 'Completed'
  6  GROUP BY PAY_METHOD
  7  ORDER BY TOTAL_REVENUE DESC;

PAY_METHOD           TOTAL_REVENUE
-------------------- -------------
Credit Card                 58300
Debit Card                  36300
UPI                          8550
Net Banking                  7000
COD                          5700
```

```
SQL> SELECT
  2       ROUND(AVG(AMOUNT_PAID), 2) AS AVERAGE_TRANSACTION_AMOUNT
  3  FROM ORDER_PAYMENTS
  4  WHERE TXN_STATUS = 'Completed';

AVERAGE_TRANSACTION_AMOUNT
--------------------------
                  14481.25
```

Report 3: Customer Payment History

Display:

- Customer name
- Order ID
- Payment mode
- Amount
- Payment status

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       O.ORDER_REF,
  4       O.ORDER_DATE,
  5       P.TXN_DATE,
  6       P.PAY_METHOD,
  7       P.AMOUNT_PAID AS AMOUNT,
  8       P.TXN_STATUS
  9  FROM CUSTOMER C
 10  JOIN ORDERS O
 11    ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  JOIN ORDER_PAYMENTS P
 13    ON O.ORDER_REF = P.ORDER_REF
 14  ORDER BY P.TXN_DATE;

CUSTOMER_NAME                  ORDER_R ORDER_DAT TXN_DATE
------------------------------ ------- --------- ---------
PAY_METHOD      AMOUNT TXN_STATUS
--------------- ------ ----------
Arjun Kumar                    ORD0001 10-JAN-26 10-JAN-26
Credit Card      55850 Completed

Priya Sharma                   ORD0002 25-JAN-26 25-JAN-26
UPI               3350 Completed

Ananya Iyer                    ORD0004 18-FEB-26 18-FEB-26
Debit Card        8300 Completed

CUSTOMER_NAME                  ORDER_R ORDER_DAT TXN_DATE
------------------------------ ------- --------- ---------
PAY_METHOD      AMOUNT TXN_STATUS
--------------- ------ ----------
Meera Krishnan                 ORD0006 15-MAR-26 15-MAR-26
Net Banking       7000 Completed

Kavya Rajan                    ORD0008 20-APR-26 20-APR-26
COD               5700 Completed

Divya Pillai                   ORD0010 18-MAY-26 18-MAY-26
UPI               5200 Completed

CUSTOMER_NAME                  ORDER_R ORDER_DAT TXN_DATE
------------------------------ ------- --------- ---------
PAY_METHOD      AMOUNT TXN_STATUS
--------------- ------ ----------
Sneha Patel                    ORD0012 20-JUN-26 20-JUN-26
Credit Card       2450 Completed

Lakshmi Venkatesan             ORD0014 15-AUG-26 14-AUG-26
Debit Card       28000 Failed

Lakshmi Venkatesan             ORD0014 15-AUG-26 15-AUG-26
Debit Card       28000 Completed
```

Payment Analysis:

```
SQL> SELECT
  2       PAY_METHOD,
  3       COUNT(*) AS TRANSACTION_COUNT,
  4       SUM(AMOUNT_PAID) AS TOTAL_REVENUE,
  5       ROUND(AVG(AMOUNT_PAID), 2) AS AVG_TRANSACTION
  6  FROM ORDER_PAYMENTS
  7  WHERE TXN_STATUS = 'Completed'
  8  GROUP BY PAY_METHOD
  9  ORDER BY TOTAL_REVENUE DESC;

PAY_METHOD           TRANSACTION_COUNT TOTAL_REVENUE AVG_TRANSACTION
-------------------- ----------------- ------------- ---------------
Credit Card                          2         58300           29150
Debit Card                           2         36300           18150
UPI                                  2          8550            4275
Net Banking                          1          7000            7000
COD                                  1          5700            5700
```